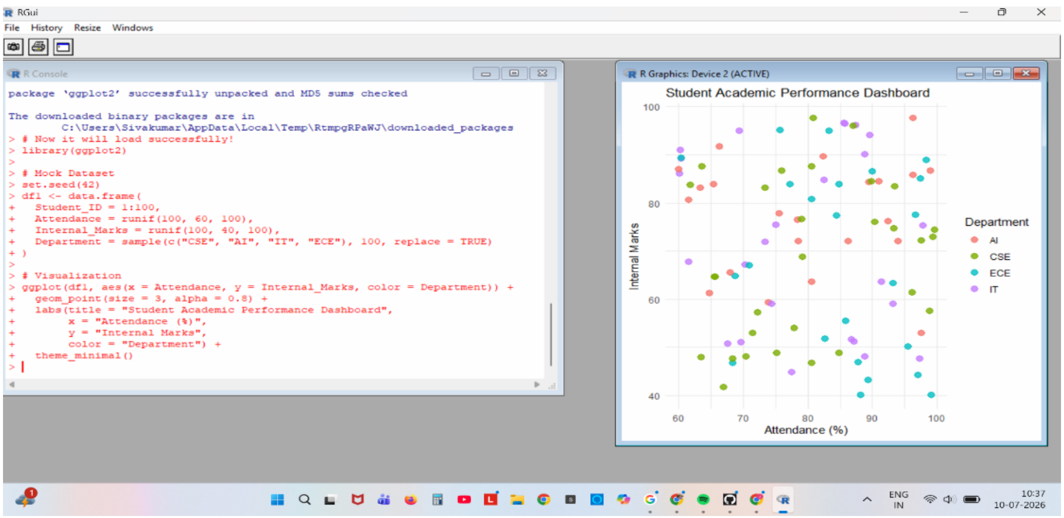
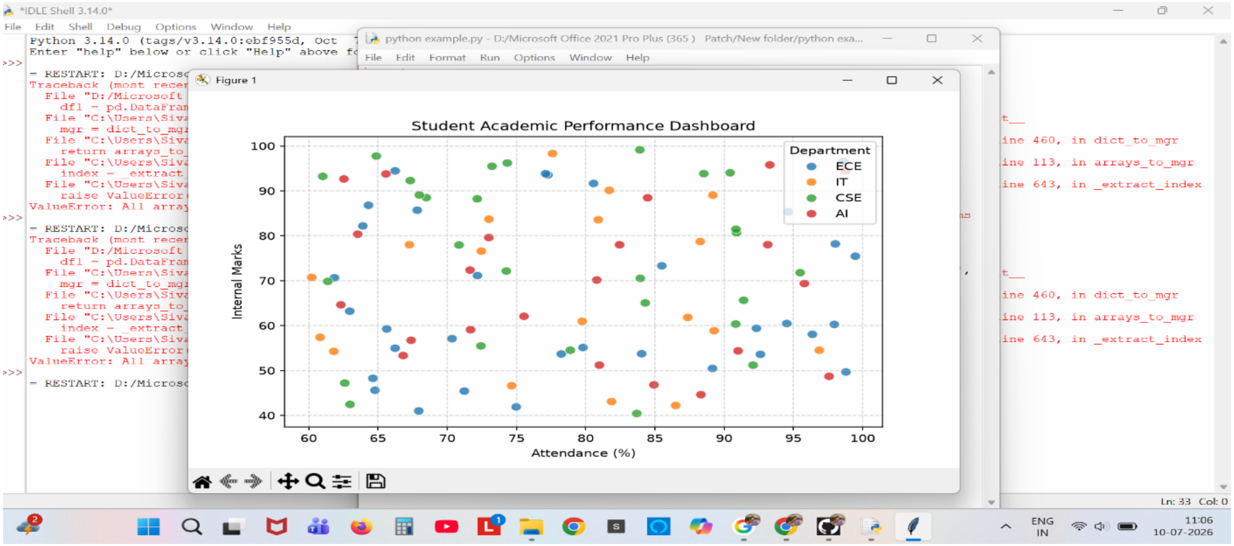
Scenario 1: Student Academic Performance Dashboard

Code:

```
        library(ggplot2)


# Mock Dataset
set.seed(42)
df1 <- data.frame(
  Student_ID = 1:100,
  Attendance = runif(100, 60, 100),
  Internal_Marks = runif(100, 40, 100),
  Department = sample(c("CSE", "AI", "IT", "ECE"), 100, replace = TRUE)
)


# Visualization
ggplot(df1, aes(x = Attendance, y = Internal_Marks, color = Department)) +
  geom_point(size = 3, alpha = 0.8) +
  labs(title = "Student Academic Performance Dashboard",
       x = "Attendance (%)",
       y = "Internal Marks",
       color = "Department") +
  theme_minimal()
```

Output:





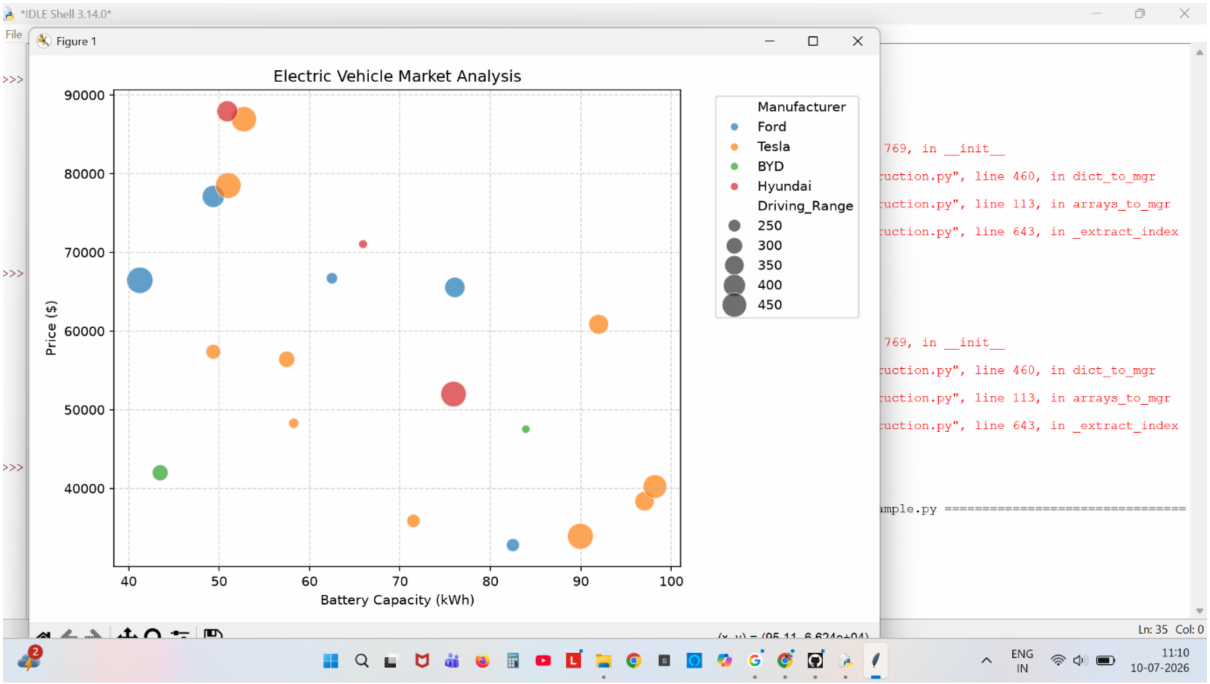

Scenario 2: Electric Vehicle Market Analysis

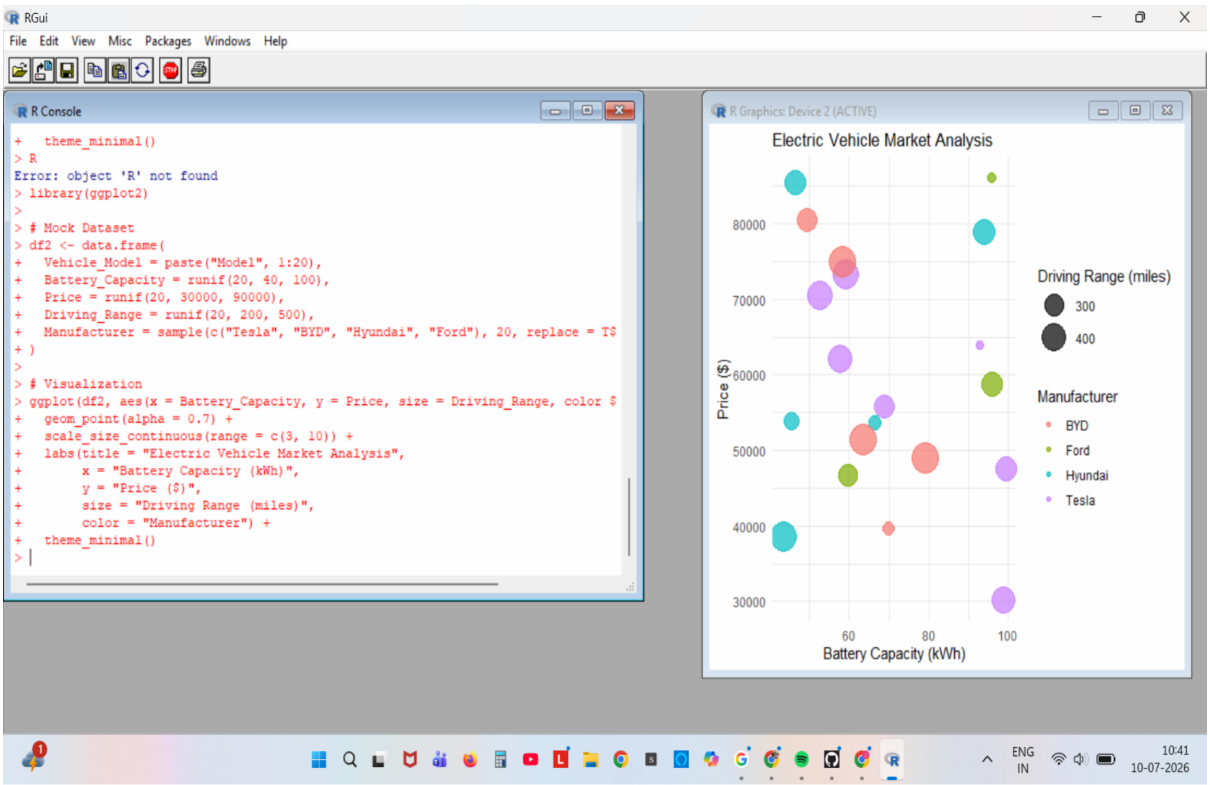
Code:

```
library(ggplot2)


# Mock Dataset
df2 <- data.frame(
  Vehicle_Model = paste("Model", 1:20),
  Battery_Capacity = runif(20, 40, 100),
  Price = runif(20, 30000, 90000),
  Driving_Range = runif(20, 200, 500),
  Manufacturer = sample(c("Tesla", "BYD", "Hyundai", "Ford"), 20, replace = TRUE)
)


# Visualization
ggplot(df2, aes(x = Battery_Capacity, y = Price, size = Driving_Range, color = Manufacturer))
+
  geom_point(alpha = 0.7) +
  scale_size_continuous(range = c(3, 10)) +
  labs(title = "Electric Vehicle Market Analysis",
       x = "Battery Capacity (kWh)",
```

```
      y = "Price ($)",
      size = "Driving Range (miles)",
      color = "Manufacturer") +
  theme_minimal()
```

Output:



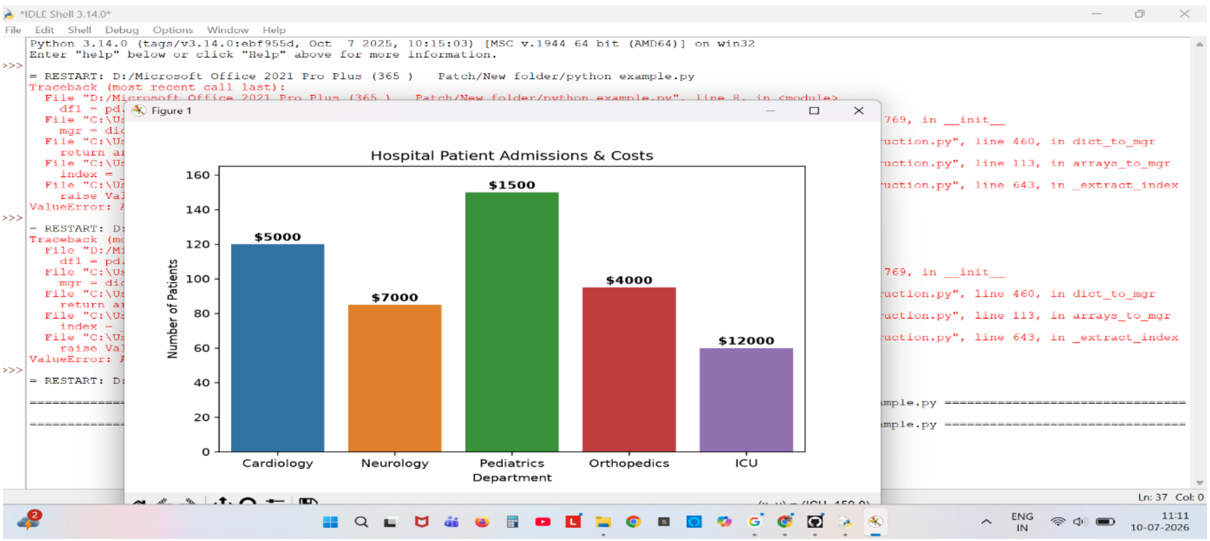



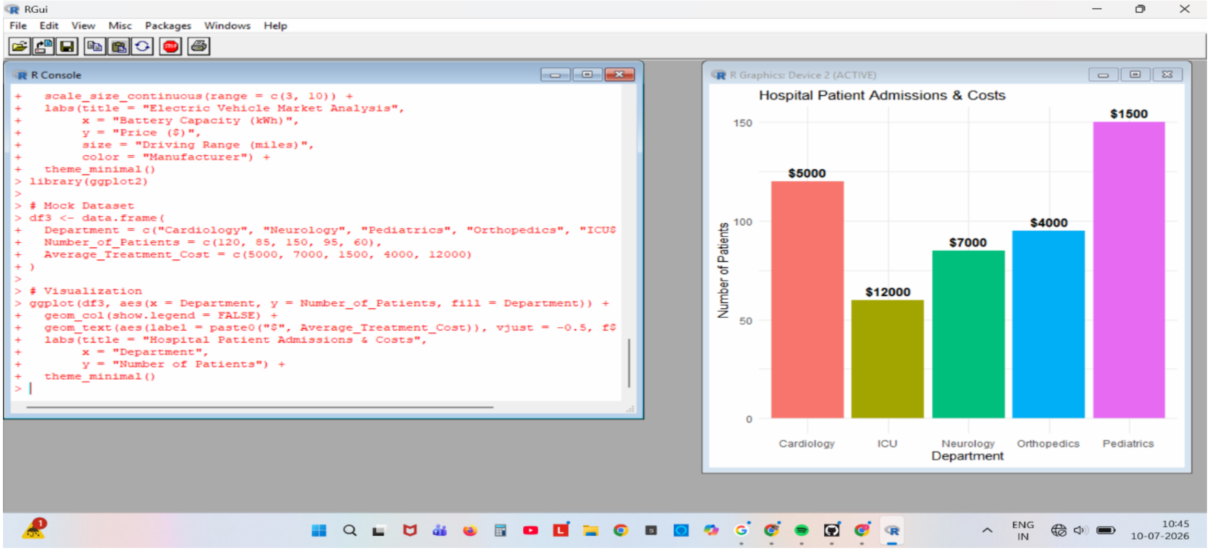
Scenario 3: Hospital Patient Admission Analysis

Code:

```
library(ggplot2)


# Mock Dataset
df3 <- data.frame(
  Department = c("Cardiology", "Neurology", "Pediatrics", "Orthopedics", "ICU"),
  Number_of_Patients = c(120, 85, 150, 95, 60),
  Average_Treatment_Cost = c(5000, 7000, 1500, 4000, 12000)
)


# Visualization
ggplot(df3, aes(x = Department, y = Number_of_Patients, fill = Department)) +
  geom_col(show.legend = FALSE) +
  geom_text(aes(label = paste0("$", Average_Treatment_Cost)), vjust = -0.5, fontface =
"bold") +
  labs(title = "Hospital Patient Admissions & Costs",
       x = "Department",
       y = "Number of Patients") +
  theme_minimal()
```

Ouput:






Scenario 4: Weather Monitoring System

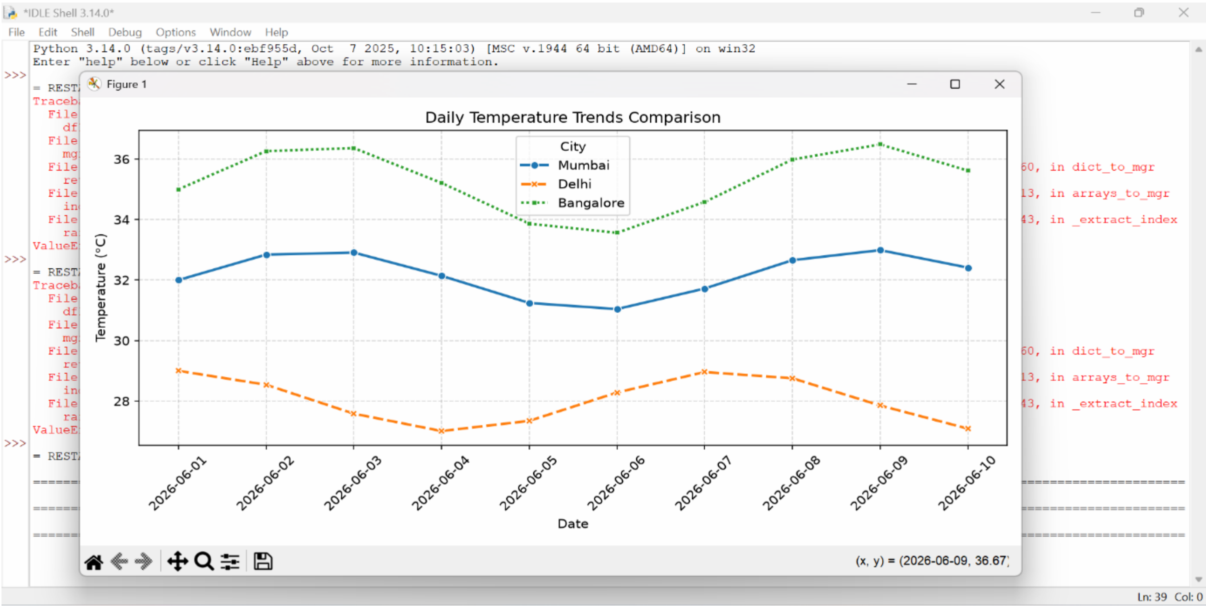
Code:

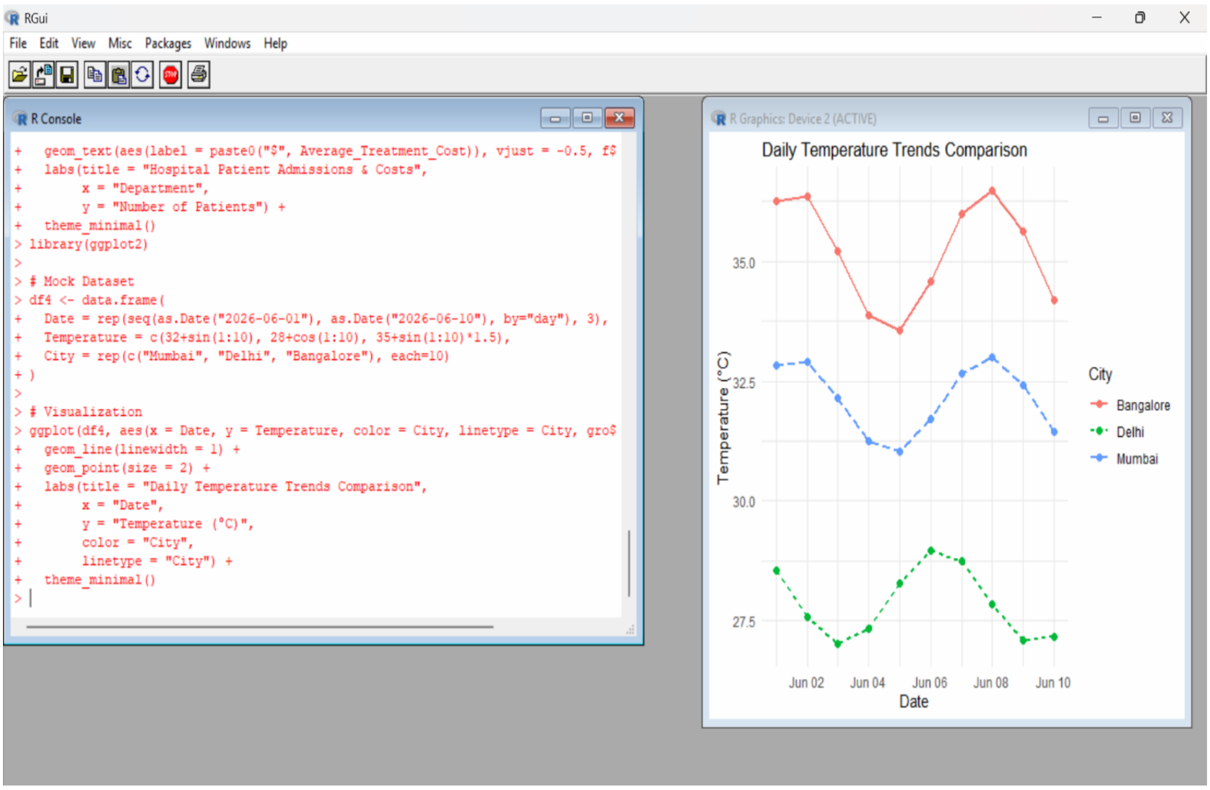
```
library(ggplot2)


# Mock Dataset
df4 <- data.frame(
  Date = rep(seq(as.Date("2026-06-01"), as.Date("2026-06-10"), by="day"), 3),
  Temperature = c(32+sin(1:10), 28+cos(1:10), 35+sin(1:10)*1.5),
  City = rep(c("Mumbai", "Delhi", "Bangalore"), each=10)
)


# Visualization
ggplot(df4, aes(x = Date, y = Temperature, color = City, linetype = City, group = City)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(title = "Daily Temperature Trends Comparison",
       x = "Date",
       y = "Temperature (°C)",
       color = "City",
```

linetype = "City") +

theme_minimal()

Output:



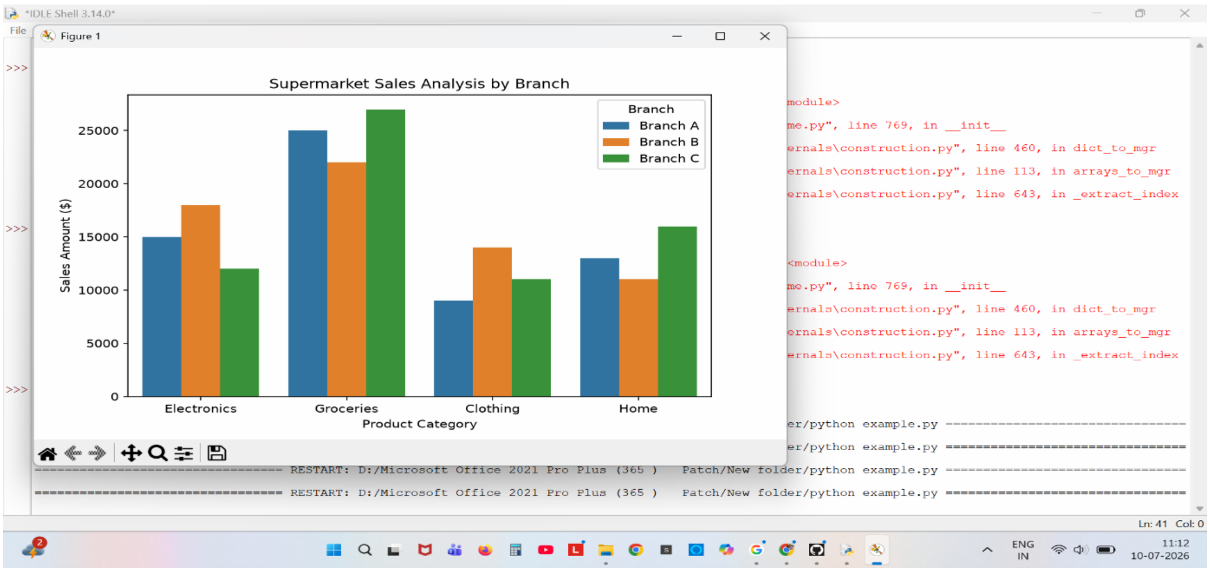



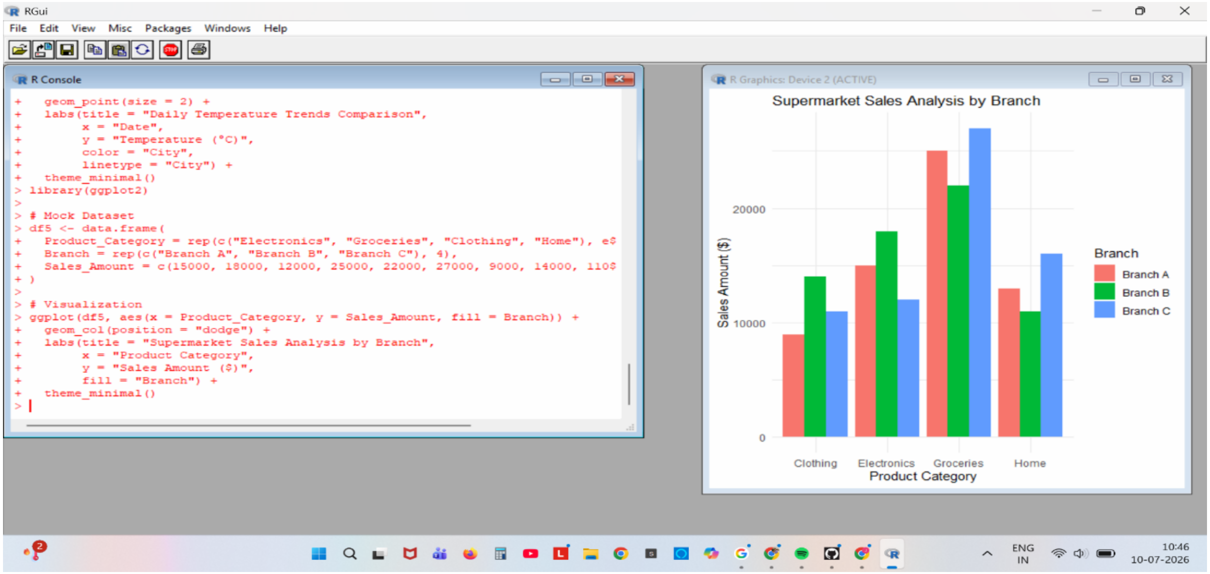
Scenario 5: Supermarket Sales Analysis

Code:

```
library(ggplot2)


# Mock Dataset

df5 <- data.frame(

  Product_Category = rep(c("Electronics", "Groceries", "Clothing", "Home"), each=3),

  Branch = rep(c("Branch A", "Branch B", "Branch C"), 4),

  Sales_Amount = c(15000, 18000, 12000, 25000, 22000, 27000, 9000, 14000, 11000, 13000, 11000, 16000)

)


# Visualization

ggplot(df5, aes(x = Product_Category, y = Sales_Amount, fill = Branch)) +

  geom_col(position = "dodge") +

  labs(title = "Supermarket Sales Analysis by Branch",

       x = "Product Category",

       y = "Sales Amount ($)",

       fill = "Branch") +

  theme_minimal()
```

Output:



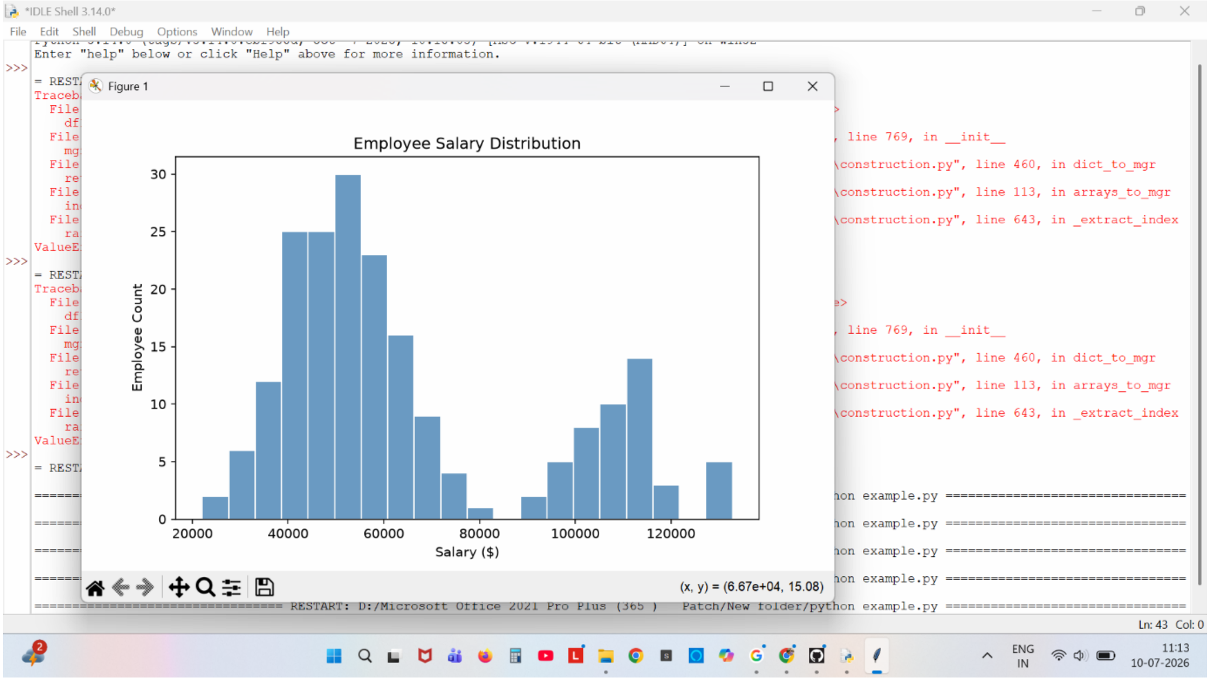


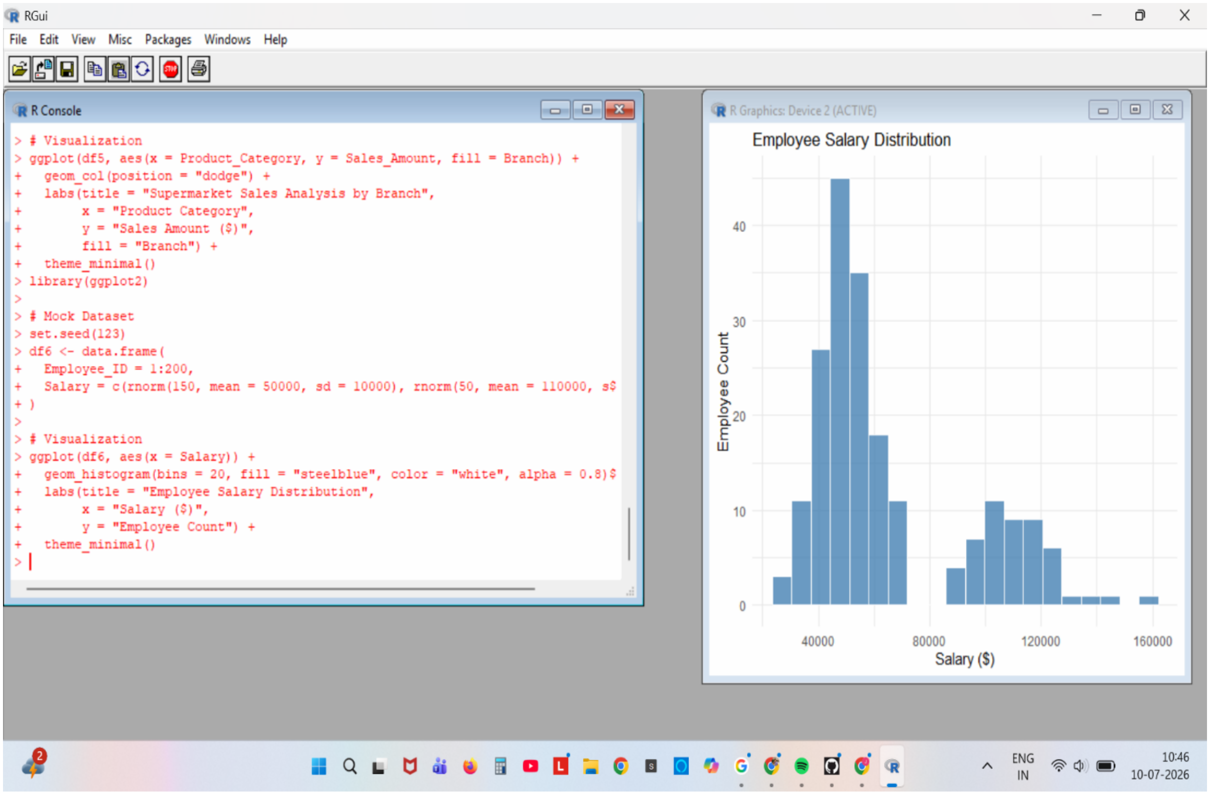

Scenario 6: Employee Salary Distribution

Code:

```
library(ggplot2)

# Mock Dataset
set.seed(123)
df6 <- data.frame(
  Employee_ID = 1:200,
  Salary = c(rnorm(150, mean = 50000, sd = 10000), rnorm(50, mean = 110000, sd = 15000))
)

# Visualization
ggplot(df6, aes(x = Salary)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white", alpha = 0.8) +
  labs(title = "Employee Salary Distribution",
       x = "Salary ($)",
       y = "Employee Count") +
  theme_minimal()
```

Output:


*IDLE Shell 3.14.0*
Enter "help" below or click "Help" above for more information.
Figure 1
Employee Salary Distribution
Employee Count
Salary ($)
(x, y) = (6.67e+04, 15.08)
Ln: 43 Col: 0

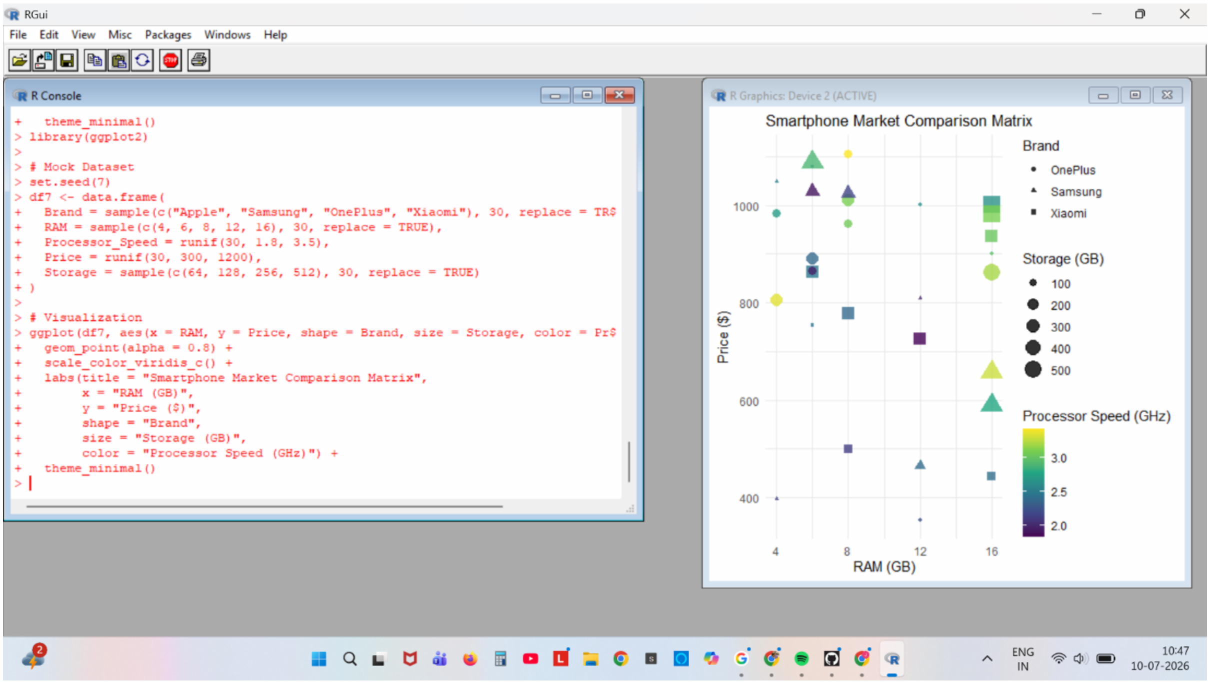

RGui
R Console
> # Visualization
> ggplot(df5, aes(x = Product_Category, y = Sales_Amount, fill = Branch)) +
+   geom_col(position = "dodge") +
+   labs(title = "Supermarket Sales Analysis by Branch",
+        x = "Product Category",
+        y = "Sales Amount ($)",
+        fill = "Branch") +
+   theme_minimal()
> library(ggplot2)
>
> # Mock Dataset
> set.seed(123)
> df6 <- data.frame(
+   Employee_ID = 1:200,
+   Salary = c(rnorm(150, mean = 50000, sd = 10000), rnorm(50, mean = 110000, s$
+ )
>
> # Visualization
> ggplot(df6, aes(x = Salary)) +
+   geom_histogram(bins = 20, fill = "steelblue", color = "white", alpha = 0.8)$
+   labs(title = "Employee Salary Distribution",
+        x = "Salary ($)",
+        y = "Employee Count") +
+   theme_minimal()
R Graphics: Device 2 (ACTIVE)
Employee Salary Distribution
Employee Count
Salary ($)

Scenario 7: Smartphone Market Comparison

Code:

```
library(ggplot2)

# Mock Dataset
set.seed(7)
df7 <- data.frame(
  Brand = sample(c("Apple", "Samsung", "OnePlus", "Xiaomi"), 30, replace = TRUE),
  RAM = sample(c(4, 6, 8, 12, 16), 30, replace = TRUE),
  Processor_Speed = runif(30, 1.8, 3.5),
  Price = runif(30, 300, 1200),
  Storage = sample(c(64, 128, 256, 512), 30, replace = TRUE)
)

# Visualization
ggplot(df7, aes(x = RAM, y = Price, shape = Brand, size = Storage, color = Processor_Speed))
+
  geom_point(alpha = 0.8) +
  scale_color_viridis_c() +
  labs(title = "Smartphone Market Comparison Matrix",
       x = "RAM (GB)",
       y = "Price ($)",
       shape = "Brand",
       size = "Storage (GB)",
       color = "Processor Speed (GHz)") +
  theme_minimal()
```

Output:

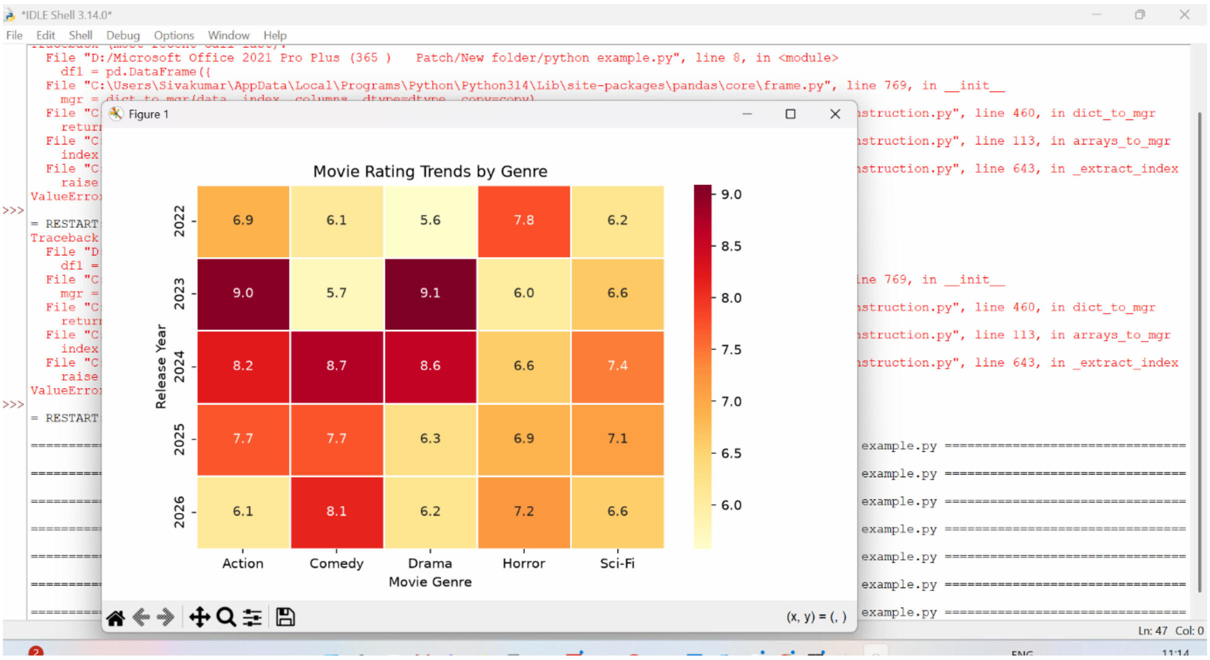

Scenario 8: Online Movie Rating Analysis

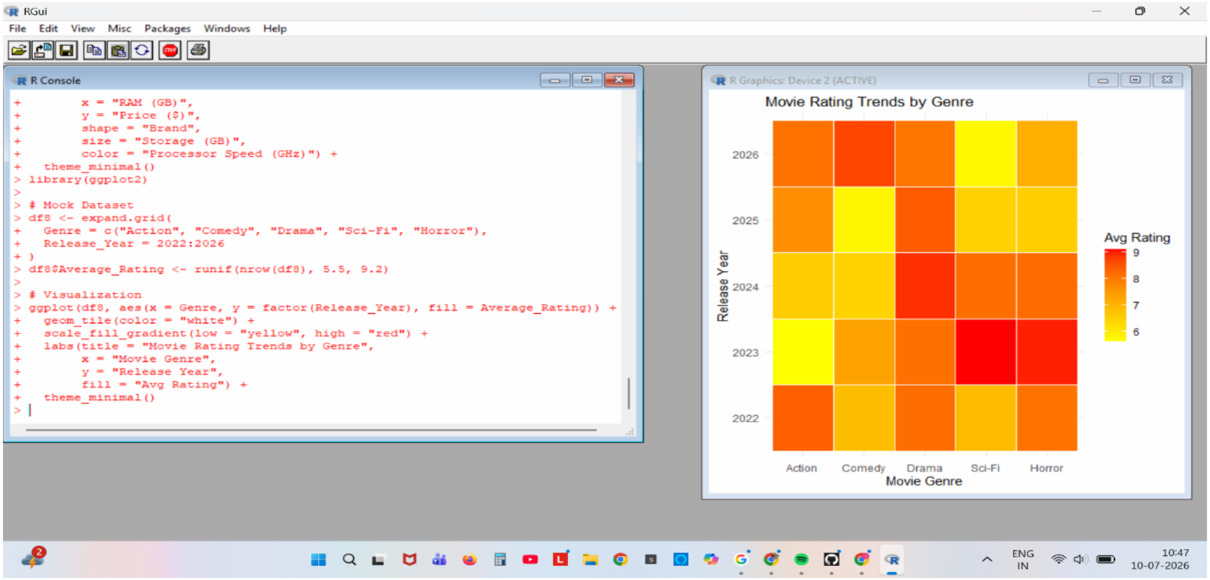
Code:

```
library(ggplot2)


# Mock Dataset

df8 <- expand.grid(
```

```
  Genre = c("Action", "Comedy", "Drama", "Sci-Fi", "Horror"),
  Release_Year = 2022:2026
)
df8$Average_Rating <- runif(nrow(df8), 5.5, 9.2)

# Visualization
ggplot(df8, aes(x = Genre, y = factor(Release_Year), fill = Average_Rating)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "yellow", high = "red") +
  labs(title = "Movie Rating Trends by Genre",
       x = "Movie Genre",
       y = "Release Year",
       fill = "Avg Rating") +
  theme_minimal()
```

Output:





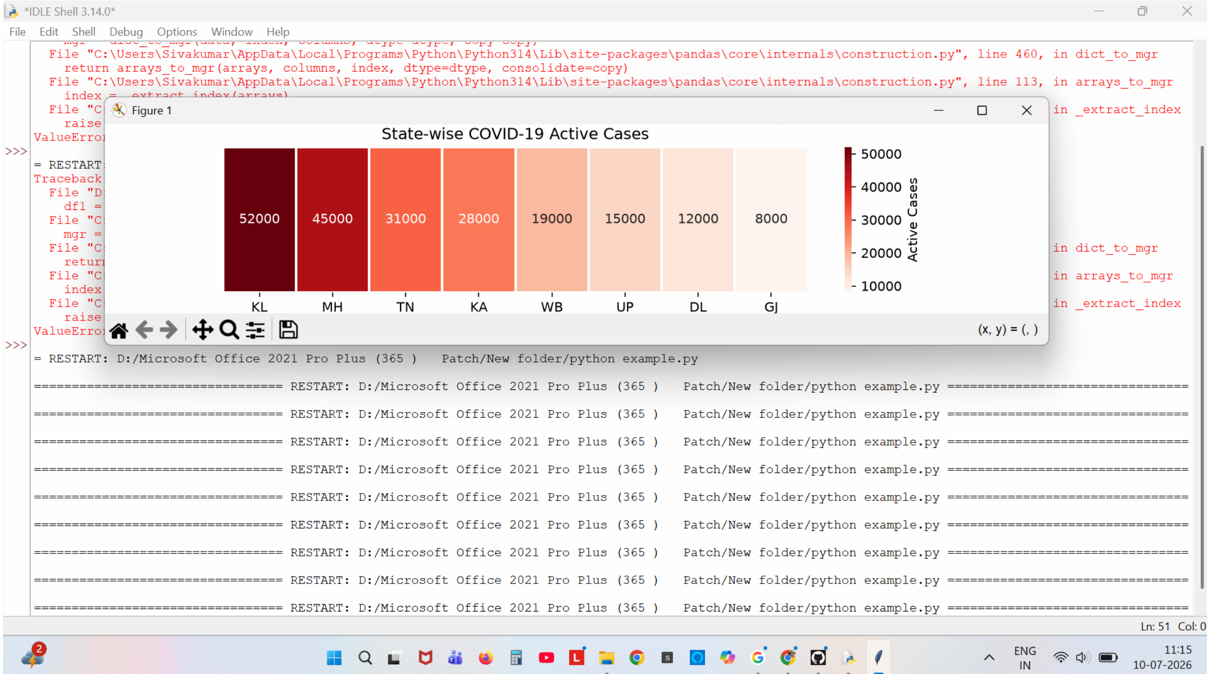

Scenario 9: COVID-19 State-wise Analysis

Code:

```
library(ggplot2)

# Mock Dataset (Indian States Sample)
df9 <- data.frame(
  State = c("MH", "DL", "KA", "TN", "KL", "UP", "WB", "GJ"),
  Active_Cases = c(45000, 12000, 28000, 31000, 52000, 15000, 19000, 8000)
)

# Visualization using a styled tile plot structure for consistency
ggplot(df9, aes(x = reorder(State, -Active_Cases), y = 1, fill = Active_Cases)) +
  geom_tile(color = "white") +
  scale_fill_distiller(palette = "Reds", direction = 1) +
  geom_text(aes(label = Active_Cases), color = "black", fontface = "bold") +
  labs(title = "State-wise COVID-19 Active Cases",
     x = "State Code",
     y = "",
     fill = "Active Cases") +
```

```
  theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(), panel.grid =
element_blank())
```

Output:

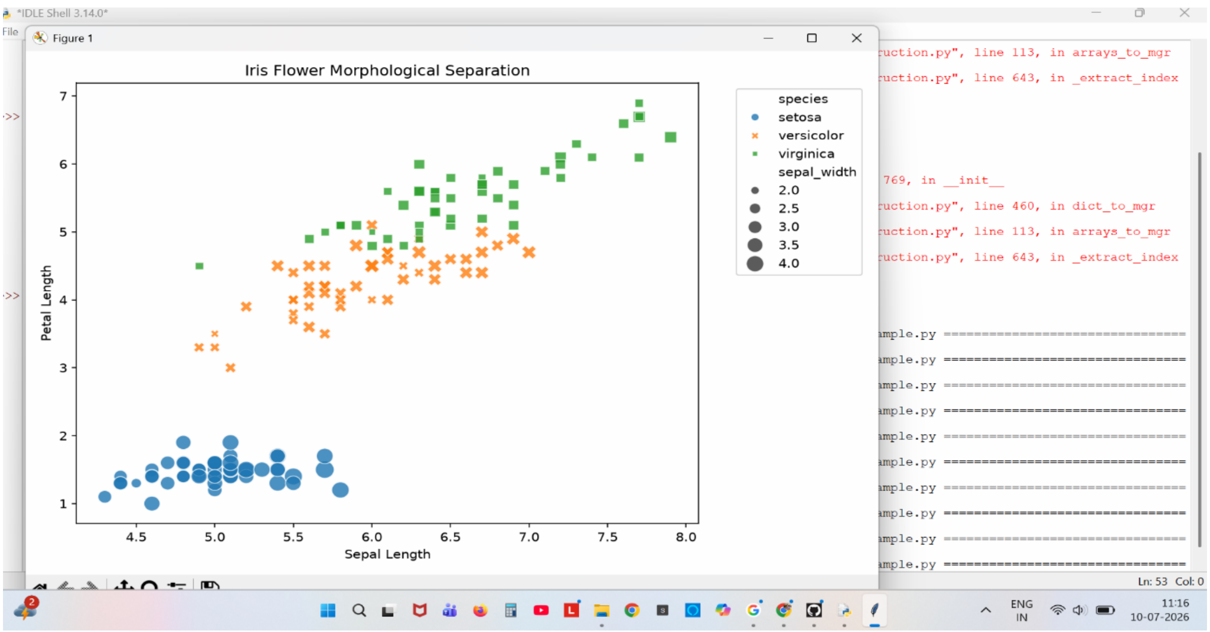


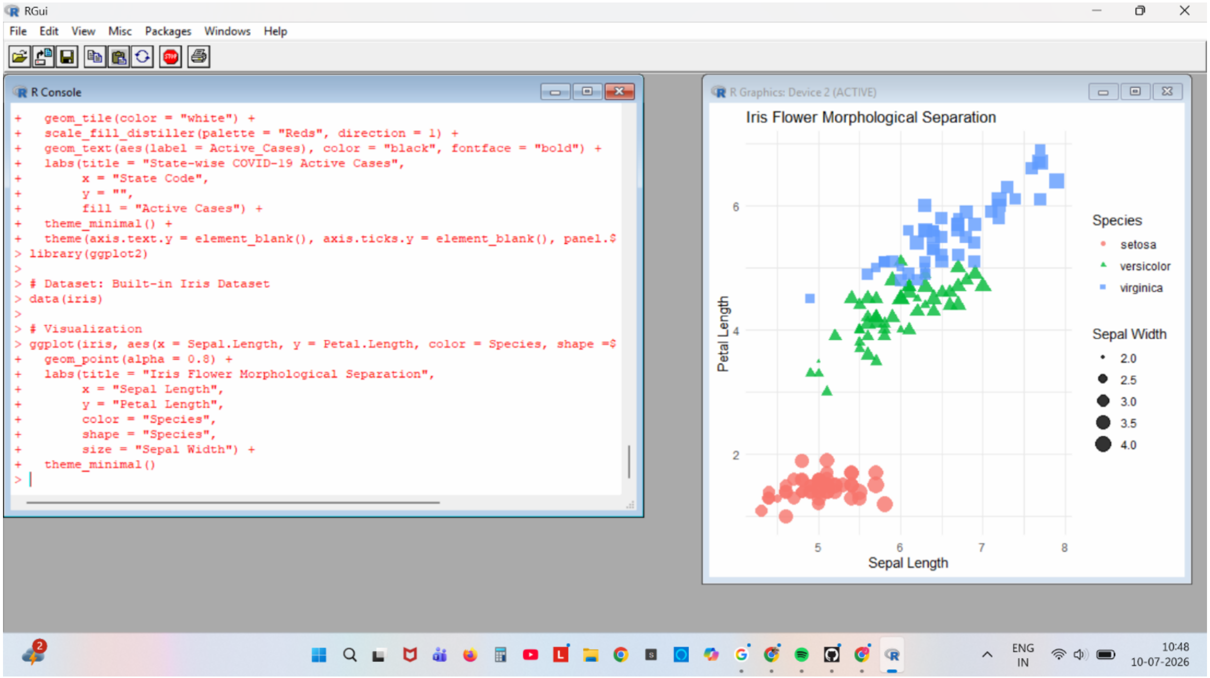

Scenario 10: Iris Flower Classification

Code:

```
library(ggplot2)


# Dataset: Built-in Iris Dataset
data(iris)


# Visualization
ggplot(iris, aes(x = Sepal.Length, y = Petal.Length, color = Species, shape = Species, size =
Sepal.Width)) +
  geom_point(alpha = 0.8) +
  labs(title = "Iris Flower Morphological Separation",
       x = "Sepal Length",
       y = "Petal Length",
       color = "Species",
       shape = "Species",
       size = "Sepal Width") +
  theme_minimal()
```

Output:




RGui
File Edit View Misc Packages Windows Help
R Console
+   geom_tile(color = "white") +
+   scale_fill_distiller(palette = "Reds", direction = 1) +
+   geom_text(aes(label = Active_Cases), color = "black", fontface = "bold") +
+   labs(title = "State-wise COVID-19 Active Cases",
+        x = "State Code",
+        y = "",
+        fill = "Active Cases") +
+   theme_minimal() +
+   theme(axis.text.y = element_blank(), axis.ticks.y = element_blank(), panel.$
> library(ggplot2)
>
> # Dataset: Built-in Iris Dataset
> data(iris)
>
> # Visualization
> ggplot(iris, aes(x = Sepal.Length, y = Petal.Length, color = Species, shape =$
+   geom_point(alpha = 0.8) +
+   labs(title = "Iris Flower Morphological Separation",
+        x = "Sepal Length",
+        y = "Petal Length",
+        color = "Species",
+        shape = "Species",
+        size = "Sepal Width") +
+   theme_minimal()
>
R Graphics: Device 2 (ACTIVE)
Iris Flower Morphological Separation
Petal Length
Sepal Length
Species
setosa
versicolor
virginica
Sepal Width
2.0
2.5
3.0
3.5
4.0
ENG
IN
10:48
10-07-2026